

Amazon Dataset: **Sales & Customer Engagement Analysis**

| Personal Project 1

| Ratings, Products, Pricings



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1. Executive Summary

This project explores how smarter pricing and merchandising strategies can boost customer engagement, conversion rates, and promotional performance in e-commerce marketplaces. Using a dataset of 1,000+ Amazon products, the analysis examines how factors like pricing, discounts, product features, and brand influence customer ratings.

After cleaning and structuring the data, several valuable insights emerged. Brand strength turned out to have a bigger impact on ratings than product details such as cable type, while steep discounts didn't always lead to better engagement. The large "Other" cable category also skewed performance results, suggesting it needs restructuring. Meanwhile, some cable types like HDMI (which were consistently rated highly) offer opportunities for premium positioning.

By optimising discounts, refining product categorisation, and highlighting strong brands, sellers could strengthen customer trust and improve product visibility. Although revenue data wasn't available, the findings point to clear opportunities for improving engagement, conversion, and promotional effectiveness through data-driven decision-making.

2. Business Problem & Objectives

This project analyses a cleaned and transformed dataset of 1,000+ Amazon products, including pricing, discounts, ratings, and product attributes, as listed on Amazon's official website and sourced from a Kaggle dataset (*Karkavelraja, 2023*). The aim is to understand how pricing strategies, discounting, product features, and brand positioning influence customer engagement, measured through ratings and rating counts.

Business Problem

Identify pricing and product feature patterns that drive higher customer engagement, and propose recommendations to optimise pricing, promotion, and product visibility on Amazon.

Key Questions

During exploration, several business questions arise:

1. How does the discount percentage impact customer engagement?

2. Are there optimal discount ranges associated with higher rating counts?
3. Do product features (e.g. fast charging, cable type) influence ratings?
4. Which brands and product categories consistently perform well?
5. Which products represent missed opportunities for promotion?

Key Performance Indicators (KPIs)

The KPIs used to address these questions can be grouped as follows:

Customer Engagement	Pricing Effectiveness	Product Performance
Average product rating	Discount percentage	Average rating by cable type
Rating count by product, brand, and category	Absolute discount (<i>money_saved</i>)	Fast-charging vs non-fast-charging engagement
–	–	Brand-level rating consistency

Assumptions and Limitations

The analysis is based on the following assumptions:

- *rating_count* is used as a proxy for customer demand and engagement.
- Higher ratings indicate greater customer trust and conversion potential.
- Sellers control pricing and discounting decisions.
- Product availability and marketing spend remain constant across products.

However, some key limitations must be acknowledged:

- Sales and revenue data are not available.
- The analysis relies on a historical snapshot of data.
- External influences such as seasonality and advertising are not considered.

3. Data Preparation & Feature Engineering

Before analysis, the raw dataset was cleaned, normalised, and transformed in Excel to ensure consistency and analytical relevance. Steps included removing redundant columns, standardising pricing and discount formats, creating derived metrics, and structuring product categories hierarchically, from broader (*category_1*) to more specific (*category_4*) levels.

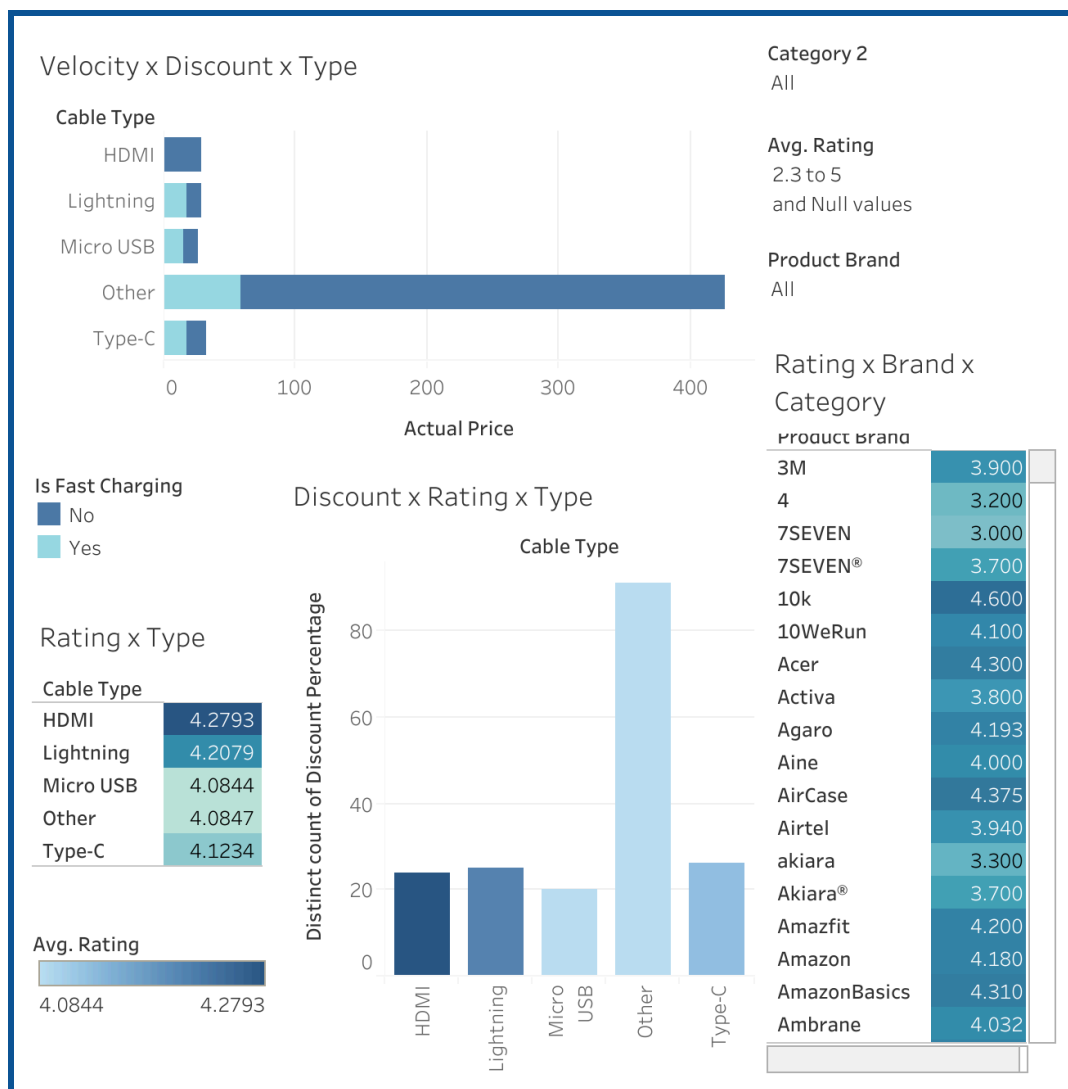
Variables Used

The final variables used are: *product_id*; *product_name*; *product_brand*; *is_fast_charging*; *cable_type*; *category_1*, *category_2*, *category_3*, *category_4*; *discounted_price*; *actual_price_manually*; *discount_percentage*; *money_saved*; *rating*; *rating_count*.

4. Analysis and Findings

As-Is State

Since most of the items purchased and rated were electronic cables, an analysis accounting for factors such as the type of cable, discounts applied and overall ratings was carried out, and is displayed on the Tableau dashboard:



Favero, Virginia. "Project 1: Amazon Product Pricing & Customer Engagement Analysis." Tableau.com, Feb. 2026.



As shown in the table *Rating x Type*, HDMI cables received the highest average rating (4.28), followed by Lightning (4.21) and Type-C (4.12). Micro-USB and other types scored slightly lower (ca. 4.08) but still demonstrated overall strong customer satisfaction, with all averages above 4.0.

In the *Discount x Rating x Type* and *Velocity x Discount x Type* charts, a large “Other” category appears to skew results. This may reflect inconsistent classification or a diverse mix of products, explaining why “Other” tends to have higher average discounts and prices than other cable types. Notably, higher discount activity in “Other” does not translate to higher ratings. The “Other” category may also be more competitive or price-sensitive, with brands relying heavily on discounts. Among the four main cable types, discount levels average around 25%, and no direct relationship is observed between discount percentage and average rating.

Data from *Velocity x Discount x Type* shows that fast-charging products appear across categories, but most revenue (reflected by darker bars) stems from non-fast-charging items. Similarly, the “Other” category contributes the most to total sales value, while fast-charging features do not significantly increase total price contribution.

Finally, the table *Rating x Brand x Category* indicates brand ratings ranging from 2.3 to 5.0. Strong performers include Redtech, Syncwire (5.0), and Acer (4.3+), while others fall closer to 3.0-3.2. This demonstrates that brand reputation and perceived quality have a stronger effect on ratings than cable type, with considerable variation among brands.

5. Business Recommendations

To-Be State

In the target state, potentially all cable products are classified into clear typologies rather than being concentrated in a broad “Other” bucket. Pricing and discounting follow consistent ranges that reflect customer engagement and perceived value instead of undifferentiated discounting. High-rated brands and cable types are marketed as top-tier, using visible ratings to justify higher prices. Fast-charging products are priced and promoted in line with the value customers actually perceive, backed by testing and clearer communication.

Gap Between As-Is and To-Be

Currently, the large, heterogeneous “Other” category distorts the bigger picture of performance and drives much of the sales value, yet its high discount levels do not translate into higher ratings. Discount patterns are inconsistent, and there is no clear link between deeper discounts and better engagement. Brand-level differences are strong, but they are not fully leveraged in visibility or promotion. Fast-charging features exist across categories but do not yet translate into a clear revenue premium. The recommendations and roadmap below are designed to close these gaps and move from the current state to the desired future state.

Actions for Future Targets

The first priority is to standardise or reclassify the “Other” cable category. It currently drives most of the total sales value and shows far greater discount variability than other types, suggesting inconsistent classification or a mix of unrelated products. A product taxonomy review is recommended to reclassify mislabeled items (for instance, USB-C adapters listed as “Other”) or to define a new cable type where appropriate. Improved classification would allow for more precise pricing strategies, clearer performance insights, and better-targeted marketing instead of broad discounting.

Secondly, high-rated cable types (particularly HDMI and Lightning) show strong demand, customer satisfaction, and a perception of quality. These categories should be positioned as premium or more reliable options, maintaining moderate discounts that support profitability without becoming too dependent on them. Although specific transaction data is unavailable, a discount range around 25-30% appears sufficient to sustain engagement without excessive price cuts. Ratings should also be highlighted within product listings to reinforce buyer confidence and justify stronger pricing power.

Further, brand reputation exerts a stronger influence on ratings than cable type, highlighting the importance of promoting high-performing brands. These should receive greater visibility across Amazon’s homepage and sponsored slots to increase conversions. Brands with excellent ratings but modest visibility may represent missed opportunities for promotion. Bundling high-rated brands with standard cable types could also enhance reach and conversion potential.

Finally, since fast-charging products do not demonstrate significantly higher revenue contribution, their positioning should be reassessed. This includes evaluating whether a pricing premium is justified, reinforcing product descriptions with clearer communication of fast-charging benefits, and testing promotional bundles (e.g. fast-charging cable +

adapter kit). If customers do not perceive additional value, repositioning could improve price-value alignment.

6. Light Implementation Roadmap

Moving from As-Is to To-Be

Short term (0–1 month): Focus on improving data accuracy and margin efficiency without major structural changes.

- Step 1: Review and adjust discount tiers by reducing excessive discounts in the “Other” category and aligning all discount levels to a 25-30% benchmark.
- Step 2: Improve product classification by reassessing “Other” items and refining tagging of cable types (HDMI, Lightning, Type-C, Micro-USB, etc.).
- Step 3: Enhance metadata accuracy by ensuring fast-charging features are correctly tagged and product naming is standardised.

Medium term (1–3 months): Validate whether margin optimisation and improved visibility drive performance gains.

- Step 1: Implement A/B testing to assess the impact of reduced discount levels in “Other” and increased visibility for high-rated brands (e.g. 4.2+).
- Step 2: Track key metrics such as average rating, rating count, and engagement velocity.
- Step 3: Evaluate discount–performance relationships by comparing test and control groups using causal inference analysis.

Long term (3+ months): Transition from static discounting to data-driven pricing and promotion strategies.

- Step 1: Integrate additional performance metrics including sales volume, conversion rate, and revenue contribution.
- Step 2: Develop dynamic pricing rules, such as reducing discounts for consistently high-rated products and increasing promotion for top-performing brands.
- Step 3: Continue refining segmentation by subdividing the “Other” category into well-structured sub-types.



7. Expected Business Impact

While precise financial figures are not available, the expected impact of moving from the current state to the target state can be outlined.

1. Optimising margins: reducing unnecessary discounting in the “Other” category could improve margins without negatively affecting customer satisfaction. Since higher discounts do not correlate with higher ratings, reducing excessive discounts could increase profitability while maintaining similar rating performance.
2. Increasing visibility for high-rated brands: adapting promotions is likely to enhance engagement and sales volume. If customers already respond positively to higher-rated brands, better placement and promotion can improve outcomes without requiring additional marketing spend.
3. Better classification accuracy: refining the “Other” category can enhance targeting efficiency, allowing for better-aligned promotional strategies and clearer performance tracking.
4. Refined fast-charging strategy: stronger communication or bundling strategies can increase their relative contribution to overall revenue. Even a small improvement in conversion rate would generate meaningful extra revenue/performance above the current level, given its presence across multiple categories.

Overall, the expected impact of this shift from the as-is to the to-be state are expected to drive improved margin efficiency, stronger brand-led growth, better promotional allocation, and clearer strategic direction through cleaner product segmentation.

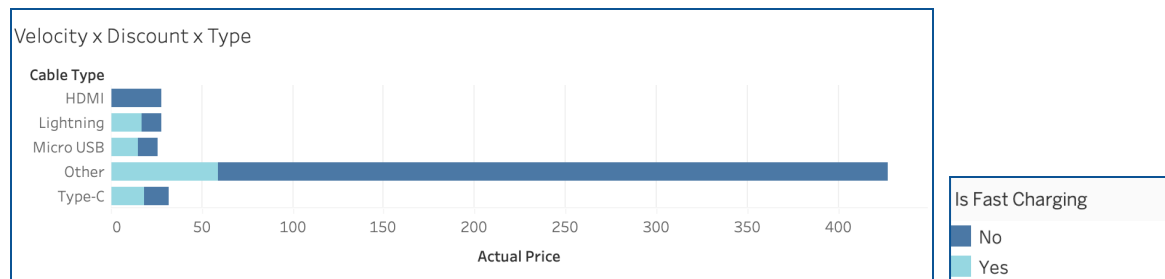


8. References and Additional Materials

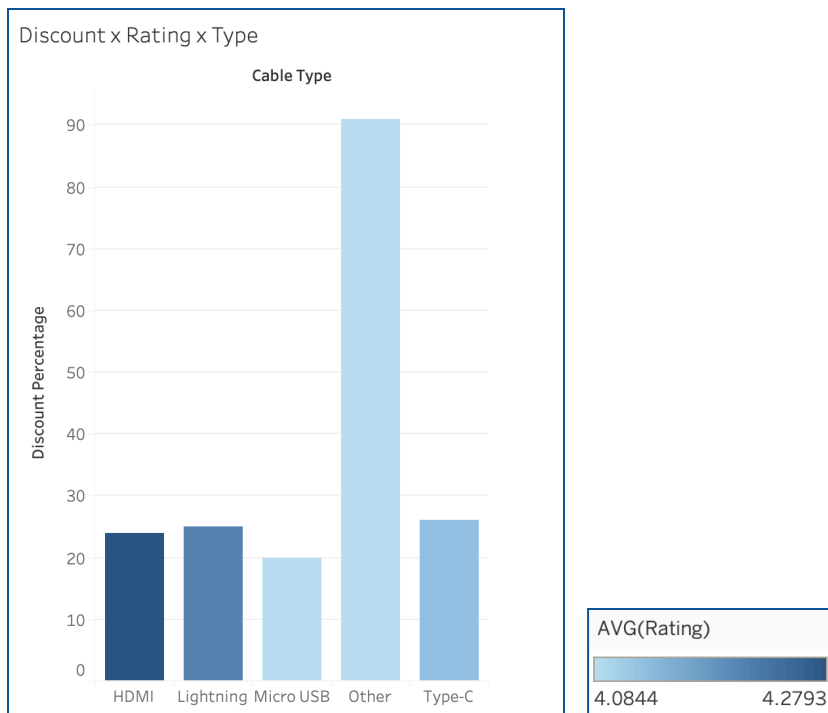
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Karkavelraja, J. "Amazon Sales Dataset." *Www.kaggle.com*, 2023, www.kaggle.com/datasets/karkavelrajaj/amazon-sales-dataset. License: CC BY-NC-SA 4.0. Accessed Feb. 2026.

Visualisation 1: *Velocity x Discount x Type*



Visualisation 2: *Discount x Rating x Type*



Visualisation 3: *Rating x Type*

Rating x Type	
Cable Type	
HDMI	4.2793
Lightning	4.2079
Micro USB	4.0844
Other	4.0847
Type-C	4.1234

Visualisation 4: *Rating x Brand x Category*

